SCALING VALUE FROM DATA

Stop calling everything P0: How to define priority for Data Assets that Executives actually respect

The Value Stream mapping framework that took one organziation from 2.5k "critic
data assets to less than 10% real P0s — and prevented millions in incidents

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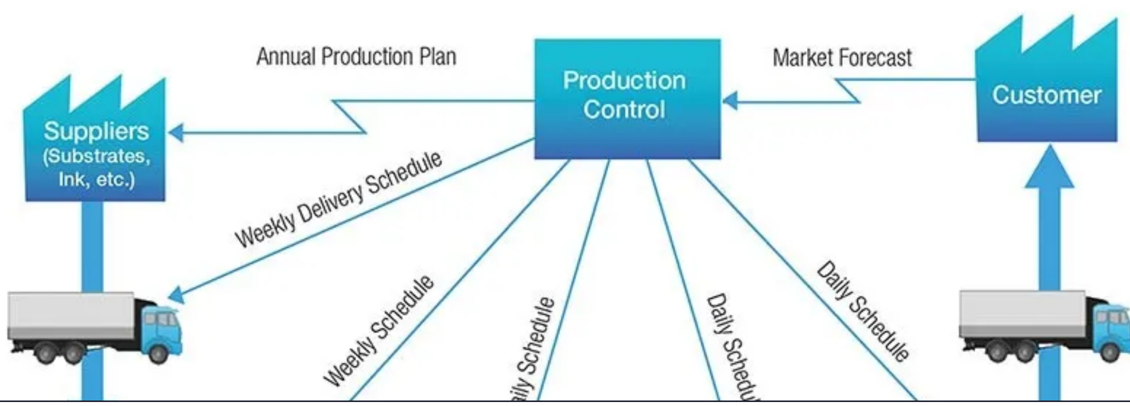
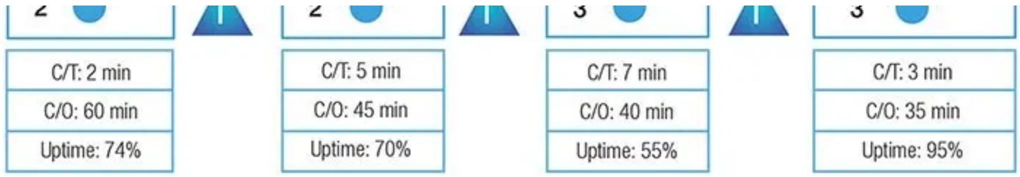
📌 *Author’s Note*

Based on P0 frameworks, I've implemented across multiple companie Methodologies are real; specific metrics and details are fictionalized for confidentiality.





A value stream maps the steps from **customer demand → customer value delivered**.

"Why is our conversion rate down 18% this month?"

The CEO's email landed at 7:15 AM. By noon, you'd found the culprit: a "deprecated" ranking table feeding your Discovery mode **Stale data for 23 days. $2.4M in lost GMV**[1]**.**

Nobody owned it. The data scientists thought analytics maintaine it. Analytics thought it was legacy. Zero alerts fired.

340,000 buyers saw the wrong product recommendations — luxury handbags replaced by men's sneakers. Conversion: 4.2% → 3.4%. Meanwhile, your team was scrambling to fix 47 "business-critical dashboards nobody opens.

This is what happens when you skip Value Stream mapping. You can protect what you can't see.

In "**Ashes on the Fire**," I promised to show you how to find your real P0s. Today, I'm delivering: the exact framework we used in previous positions to audit 2,500+ data assets and identify the only 10% that actually matter for revenue protection.

What you'll get:

- The two-criteria test that separates real P0s from executive vanity metrics
- Real-life example customer journey showing how to map critica

dependencies

- The governance framework that prevents "$2.4M stale data" disasters
- How to pitch P0 investments to your CEO (revenue protection, "data quality")

This isn't about perfection. It's about knowing which 10% assets, if they fail, cost you millions — and protecting them **before** your CEO asks.

Let's start with what P0 actually means (because most teams get it wrong).

What P0 actually means (And why most teams get it wrong)

P0 data assets must satisfy **both** conditions simultaneously:

1. **Revenue impact**: Failure directly blocks transactions, payment or revenue-generating services
2. **Customer experience impact**: Downtime degrades NPS, reviews, o core product functionality within hours

Not "important for quarterly reporting." Not "the board reviews it." Not "we'd be embarrassed if it broke."

If the answer to both questions isn't "yes within 24 hours," it's not P0.

The Critical Distinction: Data-Powered Services

Data-Powered Services are applications and microservices that consume data from the data warehouse to make real-time decisions **directly affecting customer experience or revenue.**

This includes:

- **AI & ML models** (Product Recommendations, Fraud scoring, Prici engines)
- **Reverse-ETL pipelines** feeding operational tools (CS agent interface, Curation service, Authentication service)
- **External vendor feeds & CDP integrations** (ProductsUP for SEM, Braze fetching fresh customer segments for personalized campaigns, Riskified for Fraud detection)

The litmus test: If your data doesn't power a service that **acts customers or revenue today**, it's not P0. It might be important (P1), but it's not mission-critical.

Why "The CEO Looks at It" ≠ P0

There are a set of executive dashboards showing near real-time G orders, and conversion rates. The CEO checks it every morning. T COO reviews the equivalent before leadership meetings.

Are they P0? No.

Why not? Because if the dashboard goes down for 4 hours, the marketplace still functions:

- Buyers can still purchase
- Sellers still get paid
- Fraud is still detected
- Products are still authenticated

What breaks? **Visibility**. Leadership can't see the metrics. That's annoying and politically sensitive, but it doesn't block value delivery.

Classification: P1 with high visibility (4-hour SLA, executive communication protocol).

The discipline: P0 = blocks the value stream. P1 = blocks decision making. P2 = delays reporting.

Most teams reverse this. They call executive dashboards P0 because they're politically important, then wonder why their "P0 incident response" feels like theater.

Next: Now that we know what P0 means, how do you actually find them? That's where value stream mapping comes in.

Finding P0s: Start With the Value Stream

A value stream maps every step from **customer demand → customer value delivered**.

In a factory, it's raw material → assembly → quality check → shipping. In your business, it's the complete journey from "customer lands on your site" to "customer receives value and pays you."

For an eCommerce business, you typically have two core value streams:

- **Supply/Vendor flow**: Vendor/supplier onboards → adds inventory → fulfills orders → receives payment
- **Customer/Buyer flow**: Customer discovers product → checkout →

delivery → satisfaction (or return)

Both flows are critical. No inventory = nothing to sell. No customers = no revenue. If either value stream breaks, the busin stops.

The Critical Path Principle

Any blocked step on the critical path **stops value delivery entirely**.

In data terms: if a data asset's failure blocks a critical path step within 24 hours, it's P0.

Examples of critical path blocks:

- Product catalog fails → customers see empty pages → no transactions
- Fraud detection model breaks → can't approve orders → no reve
- Payment processing data fails → vendors don't get paid → vend churn
- Inventory sync stops → oversell items that are out of stock → refunds + angry customers

Non-critical path examples:

- Executive dashboard down for 4 hours → leadership can't see metrics → annoying, but store still functions (P1)
- Weekly analytics report delayed → insights arrive Tuesday instead of Monday → delays decisions (P2)

How to Map Your Value Stream to P0 Data Assets

Here's the exact process used at a European eCommerce company to identify P0s:

Step 1: Draw Your End-to-End Customer Journey

Map both supply and customer flows with every decision point, exception path, and handoff.

Key phases to capture:

Supply/Vendor side:

- Onboarding → Catalog/Inventory Management → Order Fulfillment Payment/Settlement → Vendor Retention

Customer/Buyer side:

- Discovery → Conversion (checkout) → Delivery → Post-Purchase (satisfaction/returns) → Retention

Don't skip the exception flows: These reveal hidden dependencies (product rejected in quality checks, fraud review required, delivery failed, return processing).

Step 2: Mark the Critical Path Decision Points

Every automated decision point in your flow could block progress

Supply side:

- **Inventory sync OK?** → Can customers see available stock?
- **Quality/Compliance check passed?** → Can product go live on sit
- **Pricing data valid?** → Can checkout process orders?

Customer side:

- **Fraud Review OK/KO?** → Can buyer complete order?

- **Inventory available?** → Can we fulfill the order?
- **Payment processed?** → Can we ship the item?
- **Delivery confirmed?** → Should we trigger retention campaigns?

Why decision points matter: Every decision relies on data. If the underlying data fails, the decision can't be made → the flow stop → value is blocked.

Step 3: List the Systems Powering Each Step

For each critical path step, identify:

- The application/service (e.g., checkout service, fraud detection, inventory management, recommendation engine)
- The data source it consumes (database tables, ML model featur external APIs, CDC streams)

Example mapping:

Value Stream Step	Data-Powered Service	Data Source	Blocks Value?
Customer browses catalog	Discovery/Ranking ML	Product catalog + customer preferences	Yes - wrong products = conversion drop
Checkout - fraud check	Fraud scoring service	Transaction model + user history	Yes - can't approve/decline orders
Inventory check	Real-time inventory API	Inventory availability table	Yes - overselling = refunds + bad CX
Vendor payment	Payment settlement service	Transaction settlement data	Yes - late payments = vendor churn
Executive dashboard	BI reporting layer	Sales aggregation tables	No - reporting delay doesn't block sales

Step 4: Apply the P0 Test

For each data asset, ask three questions:

1. **Does failure block this critical path step within 24 hours?**
   - If yes → potential P0
   - If no → P1 or P2
2. **Is there a manual workaround?**
   - If yes → probably P1 (e.g., customer service can manually process returns)
   - If no → likely P0 (e.g., can't manually score 10,000 fraud checks/day)
3. **What's the revenue/compliance/NPS cost of downtime?**
   - Quantify in dollars or customer impact
   - If >$100K/day or regulatory violation → strong P0 candidat
   - If <$10K/day or just reporting delay → probably P1

Step 5: Consolidate and Get Sign-Off

At this eCommerce company, the value stream mapping exercise tool the P0 list from **2500+ data assets → ~20 candidates → 10% final P0s**.

The 20 → 10% reduction happened because:

- Some assets had manual workarounds (customer service could resolve exceptions) → moved to P1
- Some had <24hr impact window (daily batch processes) → moved P1
- Some were "politically important" but didn't block value stre → moved to P1 with exec visibility SLA

Final P0 categories (generic, safe to publish):

1. Core product/catalog data
2. Customer master data
3. Order/transaction events
4. Payment processing events
5. Fraud detection model inputs
6. Recommendation/personalization features
7. Vendor/supplier payment data
8. Inventory availability (real-time)
9. Customer segmentation (for operational tools like CDP)
0. Compliance-required data (tax, privacy, returns)

The Workshop Format

Prep (1 week before):

- Draft customer journey flows (supply + customer side)
- Invite: data leads, business stakeholders (Operations, Produc Engineering, Finance, Marketing)
- Pre-read: Share journey map + ask teams to list their "critic data assets

Workshop (2 hours):

1. **Walk the journey** (30 min): Project flows on screen, walk each step together
2. **Mark critical path** (20 min): Dot voting — where would a failu block customer value?
3. **Map data dependencies** (40 min): For each critical step, list

supporting data assets

4. **Apply P0 test** (30 min): Revenue impact? Manual workaround? Failure window?

Post-workshop (1 week):

- Consolidate P0 list with tri-party sign-off (data team, busin owner, exec sponsor)
- Define SLAs for each P0
- Document in data catalog + lineage tools

Result: One-page P0 reference that every stakeholder understood. Zero debate during incidents.

Commercial Examples: Discovery, Product Feed, Braze CDP

"Why is conversion down?" started as a Discovery ranking questio It ended as a discovery that three "independent" data assets were actually a single point of failure.

The Value Stream: Customer Acquisition → Discovery → Conversion

The Hidden Shared Dependency

Data-Powered Service	Business Impact	Team Owner
Discovery ML (personalized recommendations)	35% of conversions	Data Science
Braze CDP (marketing campaigns)	$4.2M quarterly GMV	Marketing Ops
Product Feed (Google Shopping/SEO)	$2M weekly GMV	Growth

The reality: All 3 consumed the same underlying data — product table, customer affinity signals, and catalog attributes. When o[n]e shared table went stale, all 3 degraded simultaneously.

The Cascade Failure

Day 0: Data Science migrates Discovery model to a new feature table. They deprecate the old table without checking cross-team dependencies. Nobody notices the table stops refreshing.

Week 1: Small degradations appear:

- Discovery conversion: 4.2% → 3.9% (attributed to "normal variance")
- Braze campaign CTR: 8.2% → 7.8% (blamed on "creative fatigue"
- Product Feed bounce rate: 42% → 44% (blamed on "SEO algorithm changes")

Each team sees a problem in **their** domain. Nobody connects the do[ts]

Week 2: The signals get louder:

- Discovery shows men's sneakers to customers searching for handbags (stale data from a flash sale 3 weeks ago)
- Braze sends "We miss you" emails with wrong product recommendations (stale customer behavior data)
- Google Shopping promotes sold-out items (stale inventory + popularity rankings)

Still no alarm bells. Each team troubleshoots independently.

Week 3, Monday, 7:15 AM: CEO emails VP Data and VP Product: "Why conversion down 18%?"

By noon, root cause identified: One stale table feeding 3 revenue critical services.

Total damage:

- $2.4M lost GMV (conversion decline)
- $280K degraded Braze campaigns
- $47K wasted Google Shopping ad spend

Grand total: $2.73M from one table nobody owned.

Why This Happened: The Governance Gap

Data Science checked: “Is anyone on **our team** still using this table?” Answer: No.

What they didn’t check:

- Is Marketing Ops using it for Braze segmentation?
- Is Growth using it for Product Feed rankings?
- Are there downstream dependencies documented?

The answer: No data catalog. No lineage tracking. No cross-team dependency review.

The Fix: Treating Commercial P0s as a Portfolio

Three changes prevented this from happening again:

1. Dependency Mapping
Every data asset now documents downstream consumers. The new

ranking features table shows: "Consumed by Discovery ML, Braze CI ProductsUP Feed—requires sign-off from all three teams before schema changes."

2. Shared dependency Monitoring
Real-time freshness checks + downstream impact monitoring. If Discovery conversion, Braze CTR, and Product Feed ROAS **all declin simultaneously**, automated alert triggers portfolio-level inciden response.

3. Quarterly Portfolio Review
Instead of treating Discovery, Braze, and Product Feed as independent P0s, they're now managed as the **"Commercial Revenue Engine" portfolio** with shared accountability for common dependencies.

The Lesson: Commercial P0s are interconnected

The mistake wasn't treating Discovery, Braze, or Product Feed as P0s individually. The mistake was **not understanding that they we a portfolio** sharing critical dependencies.

When customer-facing ML and personalization systems break, they rarely fail in isolation. If product ranking degrades, marketing personalization probably does too.

The discipline for commercial P0s:

1. Map the portfolio (what services share dependencies?)
2. Monitor the shared layers (not just individual services)
3. Treat failures as portfolio incidents (when one degrades, che the others immediately)

This is Executive-level thinking: you're not protecting individua
data assets. You're protecting **revenue infrastructure**.

Operation Examples: Fraud Detection, CS, and Authentication

"We can work around most data issues. But fraud detection, custor
service, and authentication? There's no manual backup when you're
operating at scale."

The Value Stream: Deliver Customer Value Safely

Operational excellence wasn't about moving fast — it was about moving **safely**. 3 data-powered services formed the backbone of trust: Fraud Detection, CS, and the Authentication Service.

Unlike commercial P0s (where failure = revenue loss), operationa
P0s have **compliance and legal stakes**: payment processor violatio
contractual SLA breaches, brand damage from counterfeit slip-through.

The 2-Hour Fraud Outage: No Manual Workaround

The fraud detection model's input data corrupted during a schema migration. Orders couldn't be scored for risk.

Immediate impact:

- 340 orders stuck in "pending review"
- Order approval rate: 420/hour → 0/hour

The impossible math:

- Manual review capacity: 50 orders/hour
- Current volume: 420 orders/hour
- Backlog grows by 370 orders/hour—manual backup impossible

The choice:
Pause risky payment methods (PayPal, BNPL) and accept $210K/hour revenue loss, or auto-approve and risk $500K+ in fraud plus payme processor penalties.

The decision: Pause payments. Accept the revenue hit.

2-hour damage:

- $87K in lost legitimate orders
- 240 abandoned carts
- 87 angry CS tickets
- $12K in apology vouchers

Total cost: $99K + brand damage.

Why P0: Zero manual alternative at eCommerce scale. When it breal revenue stops.

CS: Speed = NPS

Agents used a inhouse CS application pulling data from 6+ source: order history, payments, shipping, product status, seller reputation, and past tickets.

Efficiency gain:

- With the app: 10-min average resolution

- Without: 45+ min (manual queries across 6 systems)

When the CS tool broke for 4 hours:

- Resolution time: 10 min → 38 min
- Ticket backlog: +240%
- CSAT: 4.2/5 → 3.1/5
- Missed SLA on 47 premium seller tickets → contractual penalty risk

The lesson: At scale, CS tools aren't "nice to have." They're infrastructure. Without real-time data, agents can't meet SLAs.

The Authentication Dependency: Brand Promise on the Line

Authentication is the marketplace's value proposition (especially for trust-intensive businesses, for example Luxury fashion). The Authentication Tool used seller reputation scores to route items

- **High-reputation sellers:** Direct Shipping (fast, low cost)
- **Unknown sellers:** Warehouse Authentication (adds 3-5 days, hig cost)

What broke: Seller reputation table degraded for 11 days—scores froze.

Impact:

- 340 high-reputation items incorrectly routed to the warehouse
- Delivery delays: +3 days average
- Unnecessary warehouse costs: $4,200

- Authentication backlog: +340%

Brand damage: Premium sellers with perfect records got complaints
Trust erosion = long-term GMV risk.

Why P0: Route the wrong item to Direct Ship (skip authentication)
and a counterfeit slips through? Brand credibility destroyed.

Compliance angle: RaaS brand partners have contractual SLAs.
Routing errors = SLA breach + partnership termination risk.

The Fix: Operational P0s Require Redundancy

Fraud Detection:

- Dual-pipeline architecture (primary + backup)
- Automatic failover
- Pre-activated manual review queue

CS:

- Real-time data replication (read replica failover)
- Graceful degradation (show “last known good” data with timestamp)
- SLA monitoring (alert if resolution time >15 min)

Authentication Tool:

- Daily snapshots (failback to yesterday’s data)
- Manual override option for warehouse managers

Cost: $240K/year
Incidents prevented: $164K/year

ROI: Marginal on paper, but regulatory/compliance risk makes thi
non-negotiable.

The Lesson: Operational P0s Have Legal Stakes

Commercial P0s protect revenue. Operational P0s protect complian
contracts, and brand promises.

The discipline:

- No manual workaround at scale = P0
- Regulatory/contractual exposure = P0
- Treat as production infrastructure (redundancy, failover, graceful degradation)

The mindset shift: Operational data isn't "analytics." It's syst
of record powering business-critical services.

What to do this week (Before Part 2)

You now have the framework to identify your real P0s:

- ✅ The two-criteria test (revenue + customer impact within 24 hours)
- ✅ Value stream mapping workshop (5-step process with tri-par
sign-off)
- ✅ How commercial P0s fail as portfolios (Million worth Discovery cascade)
- ✅ How operational P0s carry legal stakes (fraud, CS, authentication)

This week, take these three actions:

For CDOs/VPs of Data

1. Run the P0 Mapping Workshop
    a. 2-hour session with Operations, Product, Engineering, Finance, Marketing
    b. Walk the customer journey together (supply + buyer flows)
    c. Apply the two-criteria test to your "critical" asset list
    d. Goal: 10 P0s maximum, 15 absolute ceiling
2. Establish Tri-Party Sign-Off
    a. Data Team: Commits to <1 hour repair, 24/7 monitoring
    b. Business Owner: Confirms criticality to value stream
    c. Executive Sponsor: Authorizes resource investment

For Existing & Aspiring Data Leaders

1. **Own a P0 End-to-End this quarter**
    a. Pick one P0 from your company's list
    b. Manage monitoring, on-call, post-mortems, and stakeholder communication
    c. Learn product ownership, not just data engineering
2. Write Post-Mortems with Business Impact
    a. ❌ Bad: “DAG failed. Added validation.”
    b. ✅ Good: “Fraud model stale for 2 hours, blocked $87K orde
    Root cause: schema change. Fix: added contracts.”

For Everyone

- **Stop Calling Everything P0:** Each P0 costs ~$45K/year (monitoring, on-call, redundancy).

  - 10 P0s protecting $18M = smart resource allocation
  - 50 P0s = diluted focus, nobody believes you

P0 isn't about perfection. It's about knowing which 10% data assets, if they fail, cost you millions—and protecting them befo your CEO sends that 7:15 AM email.

What's Next: Part 2 of the P0 Leadership Series

Finding P0s is only half the battle. Next Monday, you'll learn h to protect them.

Part 2: "How to Protect P0s: SLAs, Anti-Patterns, and the CEO Pitch"

- **How to Set SLAs Your CEO Understands**: Stop saying "99.9% uptime." Start saying "$438K annual risk."
  - You'll learn:
    - **How to translate technical SLAs to business impact**
    - **Business hours SLA weighting (peak vs off-peak vs sales seasons)**
    - **The P0 SLA scorecard framework (Acknowledgment, Time to Repair, Communication, Business Impact)**
- **Common P0 Anti-Patterns (And How to Avoid Them)**: Three ways teams sabotage their own P0 frameworks:
  - **"The CEO looks at it" ≠ P0 (visibility ≠ criticality)**
  - **Upstream dependency sprawl (when everything becomes "P0")**
  - **Over-indexing on latency (P0 ≠ real-time)**

    - Plus: How to have the conversation that preserves executiv relationships while maintaining P0 discipline.
- **The Budget Pitch That Wins**: The 4-step CEO conversation structure:
    - Show the portfolio (top 10 P0s with annual revenue impact)
    - Show past incidents ($2.77M in one year)
    - Show the investment ask (3 areas, $540K total)
    - Show the ROI (prevent one $2.4M incident = 4x return)
    - The mental math that makes budget approval obvious.

For now, run the workshop this week.

Walk your team through the Customer Journey. Apply the two-criter test. Get tri-party sign-off.

When you show up to Part 2 next Monday, you'll already have your list—ready to set SLAs, avoid anti-patterns, and pitch budget.

See you then.

Subscribe for Part 2!

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1 All metrics mentioned in this article are fictionalized for illustrative purposes.

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Debarshi Ghosh The Efficiency Playbook 7 Jan

Liked by Jimmy Pang

Really solid P0 framework. It maps neatly to financial operations, where payment, credit, and collections systems form their own “critical path” for cash flow. TCLM digs into that working ‹ and trade credit layer. Might be a useful read for data teams supporting finance.

(It’s free)- https://tradecredit.substack.com/

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